EC234AI

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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
III Semester B.E. April/May -2024 Examinations
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
ANALYSIS & DESIGN OF DIGITAL CIRCUITS WITH HDL
(2022 SCHEME)

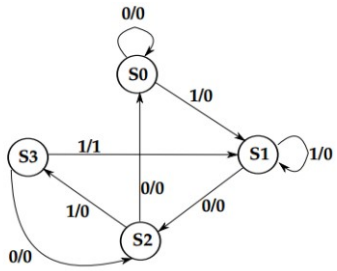
Time: 03 Hours ***Maximum Marks: 100***

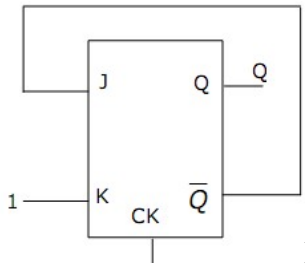
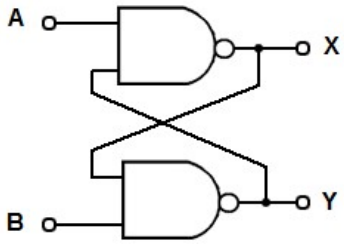
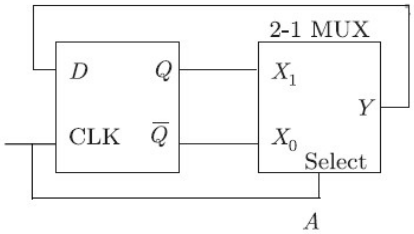
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A (Objective type for one or two marks)
(True & false and match the following questions are not permitted)

1	1.1	If the 8-bit variable "data" declared as "reg [7:0] data" is initialized to 8'b10100110, what will be its value after execution of the following code segment?
		always @(posedge clock)
		begin
		data =data << 1;
		data[0] = data[7];
		end
			1

	1.2	If reg [0:31] a_bus,, then a_bus[20-:8] is equal to_____	1
	1.3	Suppose S=8'b00010111; then S>>>2 is_____.	1
	1.4	_____ type of case statement will consider both x and z as don't care.	1
	1.5	The default value for reg data type is_____	1
	1.6	For the following Verilog module the output f equal to ____. module Q1 (f, a, b, c); input a,b,c; output f, wire t1, t2; nand G1 (t1,a,b); nor G2 (t2,b,c); or G3 (f,t1,t2); endmodule	2
	1.7	Identify the sequece detected for the FSM shown in Fig.1  Fig.1	2

	1.8	In a J-K flip flop shown below, J=Q' and K=1.Assuming that flip flop is initially cleared and then clocked for 6 pulses, the sequence at the Q output in fig1.8 will be _____  Fig1.8	2
	1.9	In figure shown A=1,B=1.The input B is now replaced by a sequence 101010…The outputs in Fig1.9,X and Y will be _____respectively(Initially the value of X=0).  Fig 1.9	2
	1.10	The sequential circuit shown in fig 1.10 consists of positive D-latch and a MUX. The latch is initially cleared (Q =0). Write the sequence of output at Y when latch is continuously clocked for 6 pulses according to the given sequence 010101.  Fig1.10	2
	1.11	The characteristic equation of JK flip flop is _____	1
	1.12	The total number of stated for a n-bit Johnson counter is _____	1
	1.13	A processor unit employs a scratch pad memory. The processor consists of 128 registers of 8 bits each. The size of the scratch pad memory is _____	1
	1.14	For the following set of micro operations, write the 16-bit control word register contents. a)R1←CrcR2 b)R3←R5 XOR R5	2

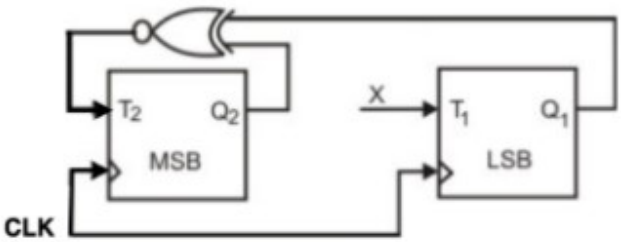
PART-B (Maximum subdivisions is limited to 3 in each question)

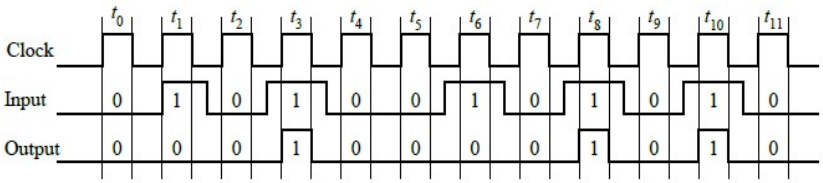
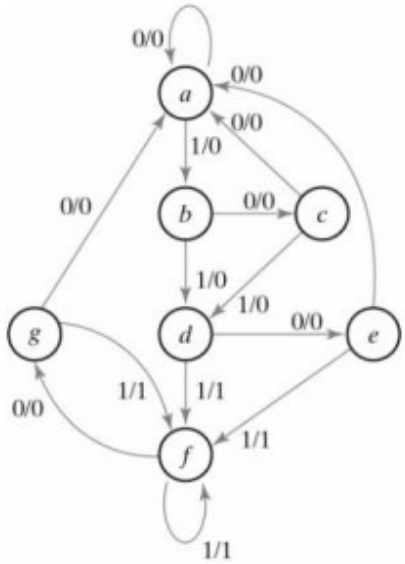
		UNIT-I	
2	a	What are the basic components of a module? Which components are mandatory?	04
	b	Write a Verilog model for a 4-bit Binary Adder & Substractor using structural	06

		modelling. Develop a test bench for the same.	
	c	Design full adder using 4:1 MUX. Write Verilog code for the same using behavioural modelling.	06

		UNIT-II	
3	a	Design and develop a 4-bit carry look ahead adder. Comment on the circuit propagation delay.	8
	b	Two decimal numbers, both between 0 and 999, need to be added to each other. Design a logic circuit that performs BCD addition of these two numbers.	8
		OR	
4	a	Design and develop a 4-bit adder/subtractor using a parallel adder. The circuit has a control input S that controls addition or subtraction operation ,S=0 results in addition and S=1 results subtraction operation.	8
	b	Design a 4bit magnitude comparator using 2-bit comparators. Write the Verilog code using structural modelling.	8

		UNIT-III	
5	a	A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively. (a) Tabulate the characteristic table. (b) Derive the characteristic equation. (c) Tabulate the excitation table. (d) Show how the PN flip-flop can be converted to a D flip-flop.	8
	b	Design a synchronous counter using T-flip flops such that it counts the sequence 1-3-5-7-9-11-13-15-1-3-5-7-9-.....and so on. Avoid Lock out condition in the design.	8
		OR	
6	a	For the state equations given, design a synchronous sequential circuit using JK- flip flops. Consider x as input and y as output. Write the state diagram and Verilog code using behavioural modelling. $A(t+1) = A(t)x+B(t)x$ $B(t+1) = A'(t)x$ $y = (A(t)+B(t))x'$	8
	b	The Fig 6b shows the partial two-bit counter using the T flip-flops which counts sequence 0-2-3-1-0.....For counting this sequence what is the expression for input X	8

		 Fig6b	

		UNIT-IV	
7	a	For the given timing diagram as shown in fig7a state the sequence which is detected and Develop a Verilog code for mealy FSM, assuming that overlapping of pattern is allowed.  Fig.7a	10
	b	Design a 4bit Ring counter using D Flip Flops.Also write any 2 differences between Johnson and ring counters.	6
		OR	
8	a	For the state diagram shown in Fig. 8a, perform state reduction with the help for state table. Write reduced state diagram. If input to this machine is sequence 01010110100 then, what is the output sequence produced by this machine?  Fig. 8a	8

	b	Draw the 4-bit universal shift register with the help of neat diagram for the following conditions.	8

S_0	S_1	Operation
0	0	No change
0	1	Shift Left
1	0	Shift Right
1	1	Parallel load

UNIT-V			
9	a	What is scratch pad memory? Explain the organization of processor unit using scratch pad memory with the help of a neat diagram.	6
	b	Design an arithmetic and logic unit with three selection variables $S_2S_1S_0$ that generate the following operations. Draw the logic diagram of one typical stage.	10

Arithmetic Operations					Logical Operations			
S_2	S_1	S_0	$C_{in}=0$	$C_{in}=1$	S_2	S_1	S_0	$C_{in}=X$
0	0	0	F=A	F=A+1	1	0	0	OR
0	0	1	F=A+B	F=A+B+1	1	0	1	XOR
0	1	0	F=A-B-1	F=A-B	1	1	0	AND
0	1	1	F=A-1	F=A	1	1	1	Complement

		OR	
10	a	Design an arithmetic circuit with one selection variables and two data inputs A and B. When s=0 the circuit has to perform the addition operation F=A+B.When s = 1 the circuit has to perform the increment operation F = A+1. Draw relevant arithmetic circuit for 4-bits.	8
	b	Explain the operation of 4-bit shifter circuit.	8

Signature of Scrutinizer: Signature of Chairman

Name: Name: